

Perturbing a bivariate-bicycle code cannot buy distance for free

An exact, perturbation-independent rate–distance obstruction for perturbed bivariate-bicycle (PBB) codes, with machine-checked certificates

Abstract

Perturbed bivariate-bicycle (PBB) codes are non-CSS stabilizer codes obtained from a CSS bivariate-bicycle (BB) parent P by adding a Z -type perturbation $[C \ D]$ to the X -check block. They have been proposed as a route past the CSS BB rate–distance envelope, on the strength of a published catalogue of 368 codes including a non-CSS $[[144, 12, 12]]$ and a $[[360, 12, \leq 24]]$ candidate.

We prove that the perturbation degree of freedom is not free. Let $\Delta = \{\lambda[C \ D] : \lambda[A; B] = 0\}$ be the *dressing space* of the perturbation. We show (i) Δ is an R -submodule of the Z -sector for $R = \mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$, (ii) the PBB dimension obeys the exact identity $k_Q = k_P - \dim \bar{\Delta}$, and (iii) any minimum-weight Z -logical of the parent that is *not* absorbed by Δ survives as a logical of Q , certifying $d_Q \leq d_Z(P)$.

Because Δ is a submodule, absorbing one minimum-weight parent logical absorbs its entire translation orbit. This upgrades the per-perturbation criterion to a **parent-level no-go theorem**: with $T(P) = \dim(M(P) + S_Z)/S_Z$, where $M(P)$ is the span of the translation orbits of *all* minimum-weight Z -logicals of P ,

$$d_Q > d_Z(P) \implies k_Q \leq k_P - T(P) \quad \text{for every perturbation } [C \ D]$$

$T(P)$ depends only on the parent and is computed *exactly* — not bounded — by a SAT enumeration that terminates in a certified UNSAT. When $T(P) = k_P$ the entire family is closed: no PBB over P retains a single logical qubit while exceeding $d_Z(P)$.

We compute T exactly for 139 of the 202 distinct parents in the published catalogue — exhaustively for every parent at $n \leq 144$. **63 parents are family-closed**, including the Gross code $A = x^3 + y + y^2$, $B = y^3 + x + x^2$ itself ($T = 12 = k_P$) and all 11 distinct catalogue $[[144, 12, 12]]$ parents. This settles 253 of the 368 catalogue rows with no per-row search at all, and it forecloses the headline direction of the construction over the most important parent in the family.

The bound is not merely valid but *exact*. A second elementary identity — the dressing space is the image of the left kernel of $[A \ B]$, whose dimension is $k_P/2$ on every BB parent — caps $\dim \bar{\Delta} \leq k_P/2$. Whenever $T(P) \geq k_P/2$, which holds on 138 of the 139 certified parents, the two bounds **sandwich**:

$$d_Q > d_Z(P) \text{ and } T(P) \geq k_P/2 \implies k_Q = k_P - T(P) \text{ exactly}$$

Perturbation does not just pay at least T logical qubits for distance; on every parent in the theorem’s class it pays *precisely* T , no more and no less. All 7 independently certified distance-increasing perturbations sit on the law exactly (slack 0, hypotheses replay-certified), and a controlled small-lattice probe — 64 sampled parents across four lattices, 26,898 valid $\delta > 0$ perturbations of Z -support ≤ 4 , distances decided by exact symplectic meet-in-the-middle — found 496 further strict increases, **every one on the law, zero above it**: instances we now know were theorem-forced rather than lucky.

On the X side the companion collapse mechanism is classified completely. The directions that *cannot* demote a parent X -stabilizer form an invariant of the single ideal $I = \text{Ann}_R(A, B)$: with I^∞ its stable power, $\dim S = 2 \dim I^\infty$, so a parent is demote-full iff I is nilpotent, demote-immune iff I is idempotent, and mixed iff neither (Theorem J-G) — reproducing the 202-parent classification in 9s with no module computation, and sharpening it to $I^2 = 0$ on all 192 demoting parents and $I^2 = I$ on all 10 immune ones. If ℓ and m are both odd, R is semisimple and no parent can demote at all: of the seven published BB instances exactly one, $[[90, 8, 10]]$ on the $(15, 3)$ lattice, is protected this way. Splitting $G = \mathbb{Z}_\ell \times \mathbb{Z}_m$ into 2-part and odd part shows exact vanishing costs two support points per occupied 2-coset, whence **no pair with** $\text{wt}(A), \text{wt}(B) \leq 3$ — **the entire family used in the literature** — **is ever mixed** (Theorem J-I), while weight 4 already realizes it: $A = (1+x)(y+y^2)$, $B = Ay$ on $(2, 3)$ is mixed with 240 of 255 classes demoting (Theorem J-J, three independent routes). An exhaustive census of 653,022,021 weight- ≤ 3 pairs over 18 lattices confirms the weight law with zero exceptions.

1 Setting and conventions

Let $R = \mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$ and let $n = 2\ell m$. A polynomial $f \in R$ acts as an $\ell m \times \ell m$ binary matrix via the commuting cyclic shifts x, y . For $A, B \in R$ the CSS **bivariate-bicycle** parent is

$$H_X = [A \ B], \quad H_Z = [B^\top \ A^\top],$$

on n qubits split into two blocks of ℓm . Commutation is $H_X H_Z^\top = AB + BA = 0$, automatic since R is commutative.

A **perturbed bivariate-bicycle** (PBB) code adds a Z -type perturbation $[C \ D]$, $C, D \in R$, to the X -check block:

$$H_Q = \left(\begin{array}{cc|cc} A & B & C & D \\ 0 & 0 & B^\top & A^\top \end{array} \right),$$

in the binary symplectic representation $(x|z)$ with symplectic product $\langle (x|z), (x'|z') \rangle_s = xz'^\top + zx'^\top$. Validity requires $M + M^\top = 0$ with $M = AC^\top + BD^\top$, i.e. **symmetry of M** , not its vanishing. (This is stated as $AC^\top + BD^\top = 0$ in some summaries; the weaker symmetric condition is the correct one and is what we verify.)

Throughout, $S_Z = \text{rowspan}(H_Z) \subseteq \mathbb{F}_2^n$ is the parent's pure- Z stabilizer space, $k_P = n - \text{rank } H_X - \text{rank } H_Z$ is the parent dimension, k_Q the PBB dimension, and

$$d_Z(P) = \min\{\text{wt}(z) : z \in \ker[A \ B] \setminus S_Z\}$$

is the parent's Z -distance. We write d_Q for the (symplectic) distance of Q and reserve *code distance* strictly for the algebraic quantity — never a decoder estimate, never a circuit or detector-graph distance.

Terminology. $C = D = 0$ recovers the CSS parent. We call a catalogue row a *reversal* when $d_Q > d_Z(P)$ is certified, and say the perturbation is *dominated* when $d_Q \leq d_Z(P)$ is certified.

2 The dressing space and the dimension identity

Define the **dressing space**

$$\Delta = \{ \lambda[C \ D] : \lambda \in \mathbb{F}_2^{\ell m}, \lambda[A; B] = 0 \} \subseteq \mathbb{F}_2^n,$$

the set of Z -parts of those first-block row combinations whose X -part cancels, and write $\bar{\Delta} = (\Delta + S_Z)/S_Z$.

Theorem G. *For every parent P and every valid perturbation $[C\ D]$:*

- (i) (Centralizer invariance.) *The pure- Z centralizer of Q equals that of P , namely $\ker[A\ B]$.*
- (ii) (Dimension identity.) *The pure- Z stabilizer group of Q is $S_Z + \Delta$, and*

$$k_Q = k_P - \dim \bar{\Delta}.$$

- (iii) (Survival criterion.) *If some minimum-weight $z \in \ker[A\ B] \setminus S_Z$ satisfies $z \notin S_Z + \Delta$, then $(0|z)$ is a nontrivial logical of Q of weight $d_Z(P)$, hence $d_Q \leq d_Z(P)$.*

Proof. (i) A vector $(0|z)$ commutes with every second-block row automatically (Z against Z). Against a first-block row $(a|c)$ the symplectic product is $a \cdot z$. Hence $(0|z)$ is in the centralizer iff $[A\ B]z = 0$ — no dependence on $[C\ D]$.

(ii) A combination λ of first-block rows is pure Z iff its X -part $\lambda[A; B]$ vanishes, in which case it contributes exactly its Z -part $\lambda[C\ D]$. Adding the second block gives pure- Z stabilizer space $S_Z + \Delta$. For the dimension, project the row space onto the X -coordinate: the image is $\text{rowspan}[A\ B]$ for both codes, while the kernel is $\{0\} \times S_Z$ for H_P and $\{0\} \times (S_Z + \Delta)$ for H_Q — the pure- Z stabilizer space, by the first part of (ii). Hence $\text{rank } H_Q = \text{rank } H_P + \dim \bar{\Delta}$, and $k = n - \text{rank } H$ gives $k_Q = k_P - \dim \bar{\Delta}$. (Note that $\text{rowspan } H_P$ is *not* contained in $\text{rowspan } H_Q$ — parent X -stabilizers are demoted — so the rank identity, not a containment, is what makes the count work.) Equivalently, via the sector identity valid for every stabilizer code, $\dim(X\text{cen}/S_X) = \dim(Z\text{cen}/S_Z) = k$: the unchanged pure- Z centralizer (i) minus the enlarged stabilizers $S_Z + \Delta$ is directly $k_P - \dim \bar{\Delta}$. (The shortcut “the X -sector of the quotient is unchanged, so only the pure- Z quotient loses dimension” is *false as an argument*: the pure- X centralizer also shrinks, $X\text{cen}(Q) = X\text{cen}(P) \cap \ker[C\ D]$ — e.g. from dimension 40 to 12 on `phase2_58` — and both the rank identity and the sector identity are what make the count work; `notes/theorem_j_xsector.md`, J.1–J.3.)

(iii) Immediate from (i) and (ii): $z \in \ker[A\ B]$ by (i), and $z \notin S_Z + \Delta$ means its class is nontrivial in the quotient of (ii). $\square$

Part (iii) is the operational content: **a single unabsorbed minimum-weight parent logical caps the PBB distance**, with no search over PBB codewords. Machine checks of (i)–(iii) on catalogue instances are in `tests/test_pbb_survival.py`; the dimension identity (ii) holds exactly on all tested catalogue rows (§8).

3 From per-perturbation to parent-level: the module structure

The step that makes a family-wide theorem possible is the following.

Lemma 1 (submodule). *Δ is an R -submodule of $\mathbb{F}_2^n$; so is S_Z , and hence so is $S_Z + \Delta$.*

Proof. The coefficient set $L = \{\lambda : \lambda[A; B] = 0\}$ is an ideal of R : if $\lambda A = \lambda B = 0$ then $(x^a y^b \lambda)A = x^a y^b (\lambda A) = 0$ and likewise for B , using commutativity of R . Since $\Delta = L \cdot [C\ D]$ and multiplication by $[C\ D]$ is R -linear, Δ is closed under the translations $x^a y^b$, i.e. is a submodule. $S_Z = \text{rowspan}[B^\top\ A^\top]$ is a submodule for the same reason. $\square$

Corollary 2 (orbit absorption). *If Δ absorbs a minimum-weight Z -logical z (i.e. $z \in S_Z + \Delta$) then it absorbs the entire translation orbit $\{x^a y^b z\}$ of z . Translations preserve weight and map $\ker[A\ B]$ to itself, so every orbit element is again a minimum-weight Z -logical.*

Now define, **from the parent alone**,

$$M(P) = \text{span} \left(\bigcup_z \text{orbit}(z) : z \text{ a minimum-weight } Z\text{-logical of } P \right), \quad T(P) = \dim \frac{M(P) + S_Z}{S_Z}.$$

Theorem H (parent-level no-go). *For every parent P and every valid perturbation $[C \ D]$,*

$$d_Q > d_Z(P) \implies k_Q \leq k_P - T(P).$$

In particular, if $T(P) = k_P$ then no perturbation of P retains a logical qubit while exceeding $d_Z(P)$: the whole family over P is closed.

Proof. Suppose $d_Q > d_Z(P)$. By Theorem G(iii), contrapositively, **every** minimum-weight Z -logical z of P must be absorbed: $z \in S_Z + \Delta$. By Corollary 2 the whole orbit of each such z is absorbed, so $M(P) \subseteq S_Z + \Delta$, whence $\bar{M} \subseteq \bar{\Delta}$ and $\dim \bar{\Delta} \geq T(P)$. Theorem G(ii) gives $k_Q = k_P - \dim \bar{\Delta} \leq k_P - T(P)$. The final sentence is the case $T = k_P$, forcing $k_Q \leq 0$. $\square$

Three features are worth emphasising. First, the hypothesis is *only* a distance increase — no assumption on C, D , on weight, or on locality. Second, $T(P)$ is a property of the parent, so one computation covers the infinite family of perturbations over P , including every $[C \ D]$ nobody has enumerated. Third, the conclusion bounds k_Q , not d_Q : the theorem does not forbid distance increase, it *prices* it.

3.1 Computing T exactly

T is obtained by a monotone SAT enumeration rather than estimated. Maintain $W = S_Z + M$, initially S_Z , and repeatedly ask a solver for $z \in \ker[A \ B]$ with $\text{wt}(z) \leq d_Z(P)$ and $z \notin W$. Nontriviality with respect to W is encoded exactly as ordinary logical nontriviality — pair z against a basis of $W^\perp$, since $\exists f \in W^\perp : f \cdot z = 1$ iff $z \notin W$. Each SAT hit is necessarily a *minimum-weight* logical (nothing outside S_Z has weight below d_Z), and its full orbit is adjoined to M . The enumeration terminates in UNSAT, which certifies that W contains every minimum-weight Z -logical; therefore the resulting T is **exact**, not a lower bound. If the budget is exhausted first, the partial T is a certified *lower* bound — the sound direction, since Theorem H only needs $\dim \bar{\Delta} \geq T$.

Two soundness details are enforced in code. (a) Every SAT witness is re-verified through two independent GF(2) paths and its weight is checked to equal the supplied $d_Z(P)$ exactly; a mismatch is raised, never absorbed. (b) A translation symmetry break (anchor support at index 0 of one block) is applied to the decision queries. This is satisfiability-preserving precisely because $\ker[A \ B]$, S_Z and every M built from full orbits are translation invariant — the solution set of each query is translation invariant, so any solution can be translated to the anchored form. The premise is machine-validated in `tests/test_translation_symmetry_break.py` (automorphism action, transitivity, and 30 differential SAT/UNSAT agreements); it buys 6.2× on the decisive UNSAT proofs (77.8 s $\rightarrow$ 12.5 s on the $\llbracket 144, 12, 12 \rrbracket$ cap-11 instance).

3.2 The trade is exact: forced saturation

Theorem H bounds k_Q from above. A second, independent observation bounds $\dim \bar{\Delta}$ from above — and for most parents the two meet.

Lemma 2 (left-kernel identity). *For every CSS BB parent, with $L = \{\lambda : \lambda A = \lambda B = 0\}$ the left kernel of $[A \ B]$, $k_P = 2 \dim L$.*

Proof. $\text{rank } H_X = \ell m - \dim L$ (L is the left nullspace of H_X). The left nullspace of $H_Z = [B^\top A^\top]$ is the transpose of $K = \ker A \cap \ker B$, so $\text{rank } H_Z = \ell m - \dim K$, and $k_P = \dim L + \dim K$. The coordinate-reversal map conjugates each cyclic shift to its transpose, so $\dim L = \dim K$. $\square$

(Verified on all 202 distinct parents, `tests/test_pbb_theorems.py`.)

Lemma 3 (dressing ceiling). $\Delta = L \cdot [C \ D]$ is the image of a linear map on L , so $\dim \bar{\Delta} \leq \dim L = k_P/2$ for every valid perturbation.

(Verified on all 368 catalogue rows via $\dim \bar{\Delta} = k_P - k_Q$.)

Theorem I (forced saturation). If $T(P) \geq k_P/2$ and $d_Q > d_Z(P)$, then $\dim \bar{\Delta} = T(P)$ and $k_Q = k_P - T(P)$ *exactly*.

Proof. Theorem H gives $\dim \bar{\Delta} \geq T$; Lemma 3 gives $\dim \bar{\Delta} \leq k_P/2 \leq T$. Squeeze, then apply Theorem G(ii). $\square$

Scope. $T \geq k_P/2$ holds on 138 of 139 certified parents. The single exception is 9a7638586033 ($n = 144$, $k_P = 12$, $T = 4$, not family-closed), where increase is only sandwiched to $4 \leq \dim \bar{\Delta} \leq 6$. On every other certified parent the exact trade law is a theorem: a distance increase pays precisely $T(P)$ logical qubits. This subsumes the empirical saturation of §5.2 — all 7 reversals have $T = k_P/2$ — and, retroactively, explains the small-lattice probe of §7: its 64 parents all have $T \in \{k_P/2, k_P\}$, so its 496 strict increases could not have landed anywhere but on the law.

4 The Gross code: its entire perturbation family is closed

The canonical Gross code (Bravyi et al. [1]) is the BB code with $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, giving $\llbracket 144, 12, 12 \rrbracket$. Running the enumeration of §3.1 on it:

Table 1: The Gross-code parent.

parent	catalogue row	ℓ, m	A	B	k_P	$d_Z(P)$	T	T exact	family closed
aebb649586c1f5ab	12_6_0194	12, 6	$x^3 + y + y^2$	$y^3 + x + x^2$	12	12	12	yes (UNSAT)	yes

$T = k_P = 12$, so by Theorem H:

Corollary 3 (Gross closure). *No perturbed bivariate-bicycle code built on the Gross code retains a single logical qubit while exceeding Z -distance 12. The entire perturbation family over the Gross code — not merely the catalogued members — is closed.*

The same holds for **all 11 distinct $\llbracket 144, 12, 12 \rrbracket$ parents present in the published catalogue**, each with $T = 12 = k_P$ certified by a terminating UNSAT, including 4c7eb964a6a3e38c, the parent of the benchmark non-CSS PBB $\llbracket 144, 12, 12 \rrbracket$ (12_6_0193, 220.4 s, 3 witnesses). That benchmark code is independently interesting — we certify it is *genuinely* non-CSS (not CSS after row operations, any qubit permutation, or any local Hadamard; a 19-parity-mask plus one-hot certificate is CP-SAT infeasible) and that its exact distance is $d = 12$, two-sided: weights ≤ 5 excluded by exact meet-in-the-middle, weight 6 by a complete classification (all 72/72 zero-syndrome vectors coincide with the stored pure- Z checks, so no weight-6 logical exists), weights 7–11 by four CaDiCaL UNSAT sector proofs (502/674/422/576 s), and a weight-12 witness verified through two independent GF(2) paths. It matches the Gross code’s $\llbracket 144, 12, 12 \rrbracket$ parameters; Corollary 3 shows that no sibling perturbation over the same parent can improve on them without giving up all 12 logical qubits.

5 Catalogue-scale results

The published catalogue [2] contains 368 PBB rows over 202 distinct CSS BB parents. Applying §3.1 to the parents:

quantity	value
distinct parents	202
parents with exact T (terminating UNSAT)	139
parents family-closed ($T = k_P$)	63
catalogue rows capped <i>a priori</i> ($k_Q > k_P - T$, no per-row work)	253 of 368
catalogue rows Theorem H permits, of the 284 classified (candidates for a genuine increase)	31

The per-length breakdown matters more than the totals, because the enumeration is **complete at every length up to $n = 144$** :

n	parents	certified	family-closed	rows	capped <i>a priori</i>
36	3	3/3	3	47	47
72	8	8/8	5	34	32
108	68	68/68	28	107	80
144	38	38/38	21	66	65
≤ 144	117	117/117	57	254	224
180	46	20/46	5	64	27
360	39	2/39	1	50	2

So over the entire catalogue at $n \leq 144$ — which contains every $\llbracket 144, 12, 12 \rrbracket$ candidate and both $\llbracket 72, \cdot, \cdot \rrbracket$ reversals — the classification is exhaustive: all 117 parents have an exact T , 57 of them are family-closed, and 224 of 254 rows are settled with no per-row search. The 63 uncertified parents are exactly the $n = 180$ and $n = 360$ ones, where a single terminating UNSAT can cost hours (the slowest completed parent took 16,493 s); that sweep continues, and because T is monotone under further enumeration these figures are lower bounds on the final closure count, never upper bounds.

5.1 A structural confinement

Cross-tabulating the 139 certified parents against their 284 catalogue rows exposes a sharp pattern. Only four ratios k_Q/k_P occur, and only three ratios T/k_P :

k_Q/k_P	$T/k_P = 1$	$T/k_P = 1/2$	$T/k_P = 1/3$	capped <i>a priori</i>
1	109	69	1	yes
5/6	11	10	—	yes
3/4	25	13	—	yes
1/2	15	31	—	15 yes / 31 no

Every row that keeps more than half of the parent’s logical qubits is capped *a priori*: $d_Q \leq d_Z(P)$ with no search. The only rows Theorem H permits to increase distance are the k -halving ones with $T/k_P = 1/2$ — exactly 31 of 284. The observed distance-increasing perturbations are confined to a k -halving class, and this is now a proved confinement rather than an empirical regularity.

5.2 The bound is saturated, not merely valid

The catalogue contains 7 perturbations for which we hold a certified reversal $d_Q > d_Z(P)$. Theorem H must permit every one of them or it is refuted. A refutation requires *both* halves — hypothesis and violated conclusion — so we verify the hypothesis independently by a replayed UNSAT lower bound on d_Q , and count any check whose hypothesis is not independently established as vacuous rather than as support.

row	n	k_P	k_Q	$d_Z(P)$	certified $d_Q >$	T	ceiling $k_P - T$	slack	hypothesis
12_6_0217	144	8	4	8	9	4	4	0	certified
phase2_58	72	8	4	4	4	4	4	0	certified
phase2_60	72	8	4	4	4	4	4	0	certified
phase2_71	108	12	6	4	5	6	6	0	certified
phase2_72	108	12	6	4	5	6	6	0	certified
phase2_88	108	4	2	4	5	2	2	0	certified
9_6_0183	108	4	2	8	9	2	2	0	certified

7/7 checked, 7/7 non-vacuous, 0 violations, and **slack exactly 0 in every case**: $k_Q = k_P - T(P)$ identically. Every certified distance increase pays exactly the price Theorem H charges, never less and never more — and Theorem I (§3.2) proves they never could: all seven have $T = k_P/2$, inside the theorem’s class, so equality was forced before it was observed.

Observation 4 (saturation — now a theorem for $T \geq k_P/2$). On every certified reversal in the published catalogue, $k_Q = k_P - T(P)$. Theorem I (§3.2) proves this is forced for every parent with $T \geq k_P/2$ — all certified parents except 9a7638586033; §7 carries the residual.

The gate is executable: `experiments/exp039_nogo_module.py gate` re-derives the table and exits nonzero on any violation, and is locked by `tests/test_pbb_nogo.py`.

6 What this does and does not settle

Settled. Over the 63 family-closed parents — including the Gross code and every catalogue $\llbracket 144, 12, 12 \rrbracket$ parent — the perturbation degree of freedom cannot produce a code that beats the parent’s Z -distance while retaining a logical qubit. This is a theorem about all $[C\ D]$, not a search result, and it is certified by terminating UNSAT proofs with hash-bound, replayable CNFs.

Priced, not forbidden. For the remaining parents, distance increase is possible but pays at least $T(P)$ logical qubits, and — by Theorem I — pays *exactly* $T(P)$ on every parent with $T \geq k_P/2$, which is every certified parent but one. It would be wrong to claim the trade is always unprofitable *relative to the parent*: on five of the seven certified reversals it is mildly profitable in kd^2/n , because $+2$ distance at $d_Z(P) = 4$ outweighs halving k (e.g. **phase2_71**: parent $12 \cdot 4^2/108 = 1.78$ versus PBB $6 \cdot 6^2/108 = 2.00$). The correct comparison is not the parent but the CSS envelope at the same length, and there the reversals lose decisively — every one of the seven is dominated at equal n by a CSS BB code with $k_C \geq k_Q$ and certified exact $d_C \geq$ the PBB’s upper bound (all seven machine-checked in `results/processed/exp036_envelope_check.json`, schema `exp036-envelope-check-v2`, against 162 certified-exact CSS candidates; the dominator shown is the strongest at that length by kd^2):

So the perturbation can beat *its own parent*, but never the best CSS code of the same length: the reversals trade a $3.0\text{--}11.1\times$ deficit against the envelope for a $\leq 1.13\times$ gain against the parent.

The X sector. The full analysis is in `notes/theorem_j_xsector.md` (J.0–J.7) and `notes/j5_attempt.md`; what the main results need is: $d_X(P) = d_Z(P)$ for every BB parent (reversal symmetry, J.0), so the Z -side framing loses no generality; the pure- X centralizer of Q shrinks by $\rho_X = \dim((R_{CD} +$

reversal	$\llbracket n, k_Q, d_Q \leq \rrbracket$	kd^2/n	dominating CSS	kd^2/n	factor
12_6_0217	$\llbracket 144, 4, 10 \rrbracket$	2.78	$\llbracket 144, 12, 12 \rrbracket$	12.00	$4.3\times$
9_6_0183	$\llbracket 108, 2, 10 \rrbracket$	1.85	$\llbracket 108, 8, 10 \rrbracket$	7.41	$4.0\times$
phase2_71, phase2_72	$\llbracket 108, 6, 6 \rrbracket$	2.00	$\llbracket 108, 8, 10 \rrbracket$	7.41	$3.7\times$
phase2_88	$\llbracket 108, 2, 6 \rrbracket$	0.67	$\llbracket 108, 8, 10 \rrbracket$	7.41	$11.1\times$
phase2_58, phase2_60	$\llbracket 72, 4, 6 \rrbracket$	2.00	$\llbracket 72, 12, 6 \rrbracket$	6.00	$3.0\times$

$S_Z)/S_Z) \geq \dim \bar{\Delta}$ (J.1); and Q 's pure- X logicals are either survivors of weight $\geq d_X(P)$ or demoted parent stabilizers paired with parent Z -logicals (J.4). Two further statements are now *decided*, one in each direction:

- *Z-distance is monotone (L1, proved).* The pure- Z centralizer is unchanged (Theorem G(i)) and $S_Z(Q) = S_Z + \Delta \supseteq S_Z$ grows, so every parent Z -logical class either survives or is absorbed — but any *surviving representative* is a parent Z -logical itself. Hence $d_Z(Q) \geq d_Z(P)$ for *every* valid perturbation, with no additional hypotheses.
- *X-distance can collapse (J.5 refuted, machine-verified).* The candidate monotonicity $d_X(Q) \geq d_X(P)$ is false: valid perturbations exist that demote a low-weight parent X -stabilizer to a genuine X -logical of Q with *no loss of logical qubits* ($\dim \bar{\Delta} = 0$). Four verified witnesses on parents with independently certified exact parent distances, each confirmed on two independent GF(2) code paths: 6_6_0099 ($n=72$: $\leq 6 < 8$), 9_6_0136 ($n=108$: $\leq 6 < 12$), 12_6_0193 ($n=144$: $\leq 6 < 12$, *the same parent as the headline non-CSS code*), 15_6_0219 ($n=180$: $\leq 6 < 14$), all four with the demoted logical of weight 6. The mechanism is the *syzygy family* $\{(C, D) : AC^T + BD^T = 0\}$: relative density $\approx 2^{-34}$ inside the validity space on the flagship parent (why earlier uniform hunts saw zero violations), a single-row demotion exists only if $M = 0$ (L3), the nontrivial excess has dimension exactly k_P on all tested parents (L4), and no Z -side mirror argument exists (L5). Earlier evidence stands with corrected interpretation: 0 collapses in 6.47M uniformly sampled perturbations is a concentration statement about the complement of the syzygy class. The collapse is *ubiquitous, not exceptional*: a syzygy-targeted census over 102 parsed parents ($n \in \{144, 180, 360\}$) finds a collapse in every bounded parent checked (49/49: all 28 at $n=144$, all 20 bounded at $n=180$, the one bounded at $n=360$); every demoted logical found has weight exactly 6; the 53 violation-free rows all lack a distance bound (exp044_n180_hunt.json). The mechanism is *computable in polynomial time* (EXP-046, Theorem J-C): the collapse family is the annihilator $c \in \ker A$, $D = 0$, with validity and partner membership both automatic, so the demotion decision is a single GF(2) rank test ($\dim \ker A \geq k_P/2$); the deterministic nullspace constructor demotes on 192/202 catalogue parents (all from single basis vectors) and collapses every bounded open-window parent but one (99/100; the exception 9_6_0175 is certified channel-free below its distance by the EXP-051 one-shot CP-SAT query: membership in $\text{rowspace}(H_X)$ at weight $\leq d_P - 1$ beside every original translated row ruled out is INFEASIBLE, so multi-row light channels below 8 are provably absent, and its single-row syzygy channel is immune by Theorem J-E exactly). EXP-048 upgrades those witnesses to exact child distances: on all 133 distance-bounded demotion parents it rebuilds the exact $D = 0$ sibling, re-verifies the parent fingerprint, $M = AC^T = 0$, $c \in \ker A$, and the pure- X weight-6 witness, and ends terminal-UNSAT after bounded descent (133/133; five require a second call at cap 3). Exact distances are 128 at 6 and 5 at 4; the strict exact parent drops are 99, of which 98 preserve dimension exactly ($\bar{\Delta} = 0$; the exception 30_6_0287 has $k = 12 \rightarrow 10$). Theorem J-E upgrades the mechanism to an *exact* characterization: with $M := \ker H_X/S_Z$ as an R -module and $L_{\text{pre}} \subset R$ the

left null space of $[A \ B]$, no single-row syzygy demotion exists iff $1 \in L_{\text{pre}} + \text{Ann}_R(M)$ — one GF(2) rank test decides collapse-hazard per parent; exhaustive class enumeration confirms it everywhere (202/202). The demote fixed set is structurally the stable image $I^\infty M$ (Theorem J-E'', proved: fixed classes are exactly the idempotent part on I -full local factors; immune $\iff IM = M$, ubiquitous $\iff I$ nilpotent on M , mixed otherwise; for $k_P > 0$ the three cases are mutually exclusive), and EXP-052 computes it in EXP-047's quotient coordinates by iterating $F_{r+1} = IF_r$ with descent-asserted termination: **on all 202 catalogue parents the demote-fraction is exactly 1 or exactly 0 (192/10, zero mixed), with chain length ≤ 3 and $I^2 M = 0$ on every demoting parent**; the immune set reproduces EXP-047's ten exactly, an independent exact enumeration over the 196 parents with $k_P \leq 20$ (2,693,100 nonzero classes, EXP-050) agrees with the algebraic predictions to the integer, and the six $k_P \in \{24, 40, 60\}$ parents are all demote-full through the algebraic route. The trichotomy is fully classified in §6.1: it is an invariant of the ideal $I = \text{Ann}_R(A, B)$ alone, mixed parents exist but never at weight ≤ 3 , and problem J-A is resolved. 9_6_0175 is X-monotone over its entire valid family ($d_X(Q) \geq 8 = d_X(P)$ for every valid $[C \ D]$): the syzygy channel is exactly empty (Theorem J-E) and the multi-row light exclusion below 8 is proved exhaustively by a one-shot CP-SAT channel query (EXP-051: membership in $\text{rowspace}(H_X)$ encoded against the dual kernel basis, weight $\leq d_P - 1$, every original row forbidden by an exclusion clause, INFESIBLE — the same certificate holds for phase2_75/76/77/83/84/87 and phase2_109 below their respective distances). The two remaining immune parents (15_6_0256, 30_6_0289) each carry an independently verified weight-8 multi-row light channel, so their X -monotonicity is organically undecided rather than merely uncertified. Practical reading: a PBB candidate can be *silently* low-distance on the X side at full k , so any downstream use needs its own two-sided distance certificate — and the audit is one GF(2) rank test away.

The caps in Theorems G/H/I do not use any of this: they cap d_Q through pure- Z survivors only, and L1 is consistent with them.

The residual class is small. The known violation of $T \geq k_P/2$ might still be the tip of a large uncharted class. It is not, at least at small lattice size: among 10,645 distinct connected $k_P \geq 2$ parents with A, B each of weight ≤ 3 (support families $2 \times 2, 2 \times 3, 3 \times 3$), exhaustive over all 43 ordered lattices with $\ell \cdot m \leq 40$ and quotiented by independent block translations with $\{A, B\}$ unordered (4,621 parents for $\ell \cdot m \leq 24$ plus 6,024 for $24 < \ell \cdot m \leq 40$; lattices 6×7 and larger skipped by an extrapolated budget model), *none* has $T < k_P/2$ — in every case computed, $T \in \{k_P/2, k_P\}$ exactly. The only known exception (9a7638586033, $k_P=12$, $T=4$) sits at $\ell \cdot m = 72$, outside this scope. Whether the residual class (parents with $T < k_P/2$) is empty beyond 9a7638586033 at larger lattices is under computation; for the known witness the residual question is *decided* (§7, EXP-045): no perturbation of it can increase distance at all.

Not settled by this paper. (i) Whether *every* $\delta > 0$ PBB is dominated by *some* CSS code — the universal envelope statement remains open (EXP-037 classifies only the 368-row published catalogue, not all codes). *Within the catalogue, every row through $n = 180$ is dominated: 254/254 at $n \leq 144$ and 64/64 at $n = 180$; only the $n = 360$ catalogue rows remain unresolved (28 candidate escapees with 22 pending; only rows with verified row records are classified; incomplete $n = 360$ CSS census, EXP-039 running).* The former $n = 180$ escape leads are decided adversely to PBB: 15_6_0224 and 15_6_0229 are both dominated by the *same* certified CSS $[[180, 8, 16]]$ (parent of 15_6_0224); the two replacement candidates ($k = 6$) were re-tightened from UB 18 to 14 and 16 (row 52's cap-16 a certified 131.7s SAT witness), hence dominated as well; and the final candidate phase2_107 — an exact PBB $[[180, 20, 6]]$ — is matched *with equality* by its own CSS parent certified exact $[[180, 20, 6]]$ (15654a1cebe3e639), so it too is dominated. No verified escape exists anywhere.

(ii) The $n = 360$ parents remain open (38 of 39 not yet certified). (iii) Everything here is a *code-level* statement. Circuit-level cost is a separate axis: our detector-error-model mechanism-distance intervals for the two $\llbracket 144, 12, 12 \rrbracket$ circuits are tied at the certified interval $[5, 12]$ on each — weight-4 exclusion completed on both circuits (perfect-matching class by meet-in-the-middle, star class by exact cover; no logical failure from any set of ≤ 4 DEM mechanisms; structural-probability DEM), with weight-12 data-logical witnesses as upper bounds (EXP-029/EXP-040) — the PBB schedule is $\approx 6\text{--}17\times$ slower on an idle machine ($p_{50} 10.3\times$, $p_{95} 16.6\times$, $p_{99} 6.3\times$), and no circuit-level result has favoured the PBB candidate. **Provisional matched-cell Monte Carlo (EXP-040, 2026-08-19):** the first completed matched schedule pair at $p = 0.002$ separates clearly — CSS Gross sched#4382: block LER 6.33×10^{-3} (95 % simultaneous CI $(5.39, 7.37) \times 10^{-3}$, 52,000 shots) versus PBB sched#4107: 2.21×10^{-2} (CI $(2.112, 2.311) \times 10^{-2}$, 172,000 shots) — disjoint intervals, the PBB $\approx 3.5\times$ higher, with cumulative decode-core time per shot $\approx 10\times$ higher (throughput proxy only; no pooled latency quantiles claimed). One finished cell of a 10-cell grid: a provisional single-pair data point, not a verdict. A verified one-ancilla mixed-stabilizer syndrome-extraction circuit for the non-CSS PBB, benchmarked against the Gross code under identical noise, remains the decisive end-to-end experiment.

Honest limitations. T is exact only where the enumeration terminated in UNSAT (139/202 parents; all 117 at $n \leq 144$); elsewhere it is a certified lower bound, which is the sound direction. $d_Z(P)$ is used as supplied and is independently certified for the parents in the table. Catalogue-reported distances are treated as untrusted throughout; where we quote a distance we quote our own two-sided certificate, and one catalogue heuristic bound was found to be $\geq 2.5\times$ loose ($d \leq 40$ advertised, ≤ 16 re-verified).

6.1 The demote law is an ideal invariant, and no trinomial pair is mixed

The demote mechanism above is decided by a module computation ($S = I^\infty M$ inside $M = \ker H_X/S_Z$). It is in fact governed by the ideal alone. Decompose $R = \prod_\beta R_\beta$ into local Artinian factors and recall $I = L_{\text{pre}} = \text{Ann}_R((a, b))$ — *not* the module annihilator $\text{Ann}_R(M)$ that also appears in Theorem J-E. The two are not interchangeable: I is full exactly where $a_\beta = b_\beta = 0$ and zero where either polynomial is a unit, whereas $\text{Ann}_R(M) = (\bar{a}, \bar{b})$ — H_X acts through transposed circulants — is full exactly on the unit factors, dual to I with $\dim I + \dim \text{Ann}_R(M) = \ell m$.

Theorem J-G (ideal invariance). $\dim S = 2 \dim_{\mathbb{F}_2} I^\infty$, and for $k_P > 0$

$$\text{demote-full} \iff I \text{ nilpotent}, \quad \text{immune} \iff I^2 = I, \quad \text{mixed} \iff 0 \neq I^\infty \neq I,$$

with $M = 0 \iff I = 0$. *Proof.* $I = \prod I_\beta$ and $M = \bigoplus M_\beta$ split componentwise; $I_\beta = R_\beta$ exactly where $A_\beta = B_\beta = 0$, and there the localized H_X vanishes, so $M_\beta = R_\beta^2$; proper $I_\beta \subseteq \mathfrak{m}_\beta$ is nilpotent, so $I^\infty = e_I R$. Theorem J-E'' gives $S = e_I M = \bigoplus_{\text{full}} M_\beta$ of dimension $2 \dim e_I R$. For the immune case, $R = \mathbb{F}_2[G]$ and each of its blocks is Frobenius, so $\dim I_\beta = \dim R_\beta / (A_\beta, B_\beta) = \frac{1}{2} \dim M_\beta$ and $M_\beta = 0 \iff I_\beta = 0$. $\square$

Machine consequences (EXP-053 classifies the whole catalogue in 8.9s, replacing the module chains of EXP-052; the dual-ideal audit below adds ≈ 4 minutes): the ideal route reproduces EXP-052 on **all** 202 parents with zero mismatches, and sharpens it — $I^2 = 0$ on all 192 demoting parents (nilpotency index exactly 2) and $I^2 = I$ on all 10 immune ones, so the catalogue realizes only the two extremes. The dual ideal is audited independently: $\text{Ann}_R(M) = (\bar{a}, \bar{b})$ on 202/202 parents by two unrelated computations (the quotient action on M versus the ideal generated by the two H_Z blocks), with $\dim I + \dim \text{Ann}_R(M) = \ell m$ everywhere; the bar is essential, since (a, b) is bar-invariant on only

156 of the 202 parents. The two annihilators are consequently never interchangeable — substituting $\text{Ann}_R(M)$ would return $2(\ell m - k_P/2)$ (100, 172, 344 across the ten immune parents) instead of the correct $\dim S = k_P \in \{8, 16\}$, and would misclassify the mixed witness as demote-full.

Corollary J-G1 (odd lattices). If ℓ, m are both odd then $x^\ell - 1, y^m - 1$ are squarefree, R is semisimple, every ideal is idempotent, and *no* parent admits a single-row syzygy demotion: the collapse mechanism requires an even lattice dimension (3,600 parents over nine odd lattices, zero violations). Of the seven published BB instances exactly one is protected this way — $\llbracket 90, 8, 10 \rrbracket$ on $(\ell, m) = (15, 3)$ is immune, while $\llbracket 72, 12, 6 \rrbracket$, $\llbracket 108, 8, 10 \rrbracket$, the Gross $\llbracket 144, 12, 12 \rrbracket$, $\llbracket 288, 12, 18 \rrbracket$, $\llbracket 360, 12, \leq 24 \rrbracket$ and $\llbracket 756, 16, \leq 34 \rrbracket$ all have $I^2 = 0$, i.e. every nonzero class demotes.

Split $G = \mathbb{Z}_\ell \times \mathbb{Z}_m = G_2 \times G_{\text{odd}}$ into 2-part and odd part. Then $\mathbb{F}_2[G_{\text{odd}}] = \prod_\chi F_\chi$ is semisimple and $\mathbb{F}_2[G_2]$ is local with nilpotent maximal ideal, so the local factors of R are $R_\chi = F_\chi[G_2]$, in which G_2 is an F_χ -basis.

Theorem J-H (exact-vanishing coset law). Partition the support of $a \in R$ by 2-part, $S_h = \{u \in G_{\text{odd}} : hu \in \text{supp } a\}$. Then $a_\chi = \sum_{h \in G_2} (\sum_{u \in S_h} \chi(u))h$; hence $a_\chi = 0$ iff every coset sum vanishes, a_χ is a non-unit iff the total sum vanishes, and $a_\chi = 0$ forces $|S_h| \geq 2$ on every occupied coset. A polynomial of weight ≤ 3 that vanishes exactly somewhere therefore occupies a single coset, and is a *scalar* — zero or unit — at every factor.

Theorem J-I (weight law). If $\text{wt}(A) \leq 3$ and $\text{wt}(B) \leq 3$ then (A, B) is **not mixed**, on any lattice: mixed needs a factor where both vanish (forcing single cosets, hence zero-or-unit everywhere) and a factor where both are non-units but not both zero, which is then impossible. A mixed parent requires weight ≥ 4 spread over ≥ 2 cosets in one of A, B .

This explains the all-or-nothing law without enumeration: the published catalogue is weight- $(3, 3)$. Confirmations: an exhaustive census of 653,022,021 translation-normalised weight- ≤ 3 pairs over 18 lattices — every catalogue lattice, the previously underexplored $(12, 12)$ and $(15, 12)$, and a parity grid — finds **zero** mixed pairs (EXP-054, with 450 pairs re-decided by an independent route, zero disagreements); and the single-coset predicate separates the catalogue *exactly*: all 10 immune parents are single-coset, all 192 demoting parents are not. For $m = 6$ this reads off by eye: immunity iff all A -monomials share one y -parity and all B -monomials share one y -parity.

Theorem J-J (mixed exists; the bound 4 is attained). On $(\ell, m) = (2, 3)$, $A = (1+x)(y+y^2)$ and $B = Ay$ give $\dim I = 4$, $\dim I^2 = \dim I^\infty = 2$: a **mixed** parent with $k_P = 8$, $\dim S = 4$ and exactly $2^8 - 2^4 = 240$ of 255 nonzero classes demoting — verified by the ideal chain, by the EXP-052 module chain $[8, 4, 4]$, and by brute-force evaluation of all 255 classes through the EXP-047 rank test. Problem J-A is resolved affirmatively, while Theorem J-I shows mixed can never occur inside the trinomial family. Open: the sharp weight/lattice map at weight ≥ 4 , and whether any mixed parent has useful $\llbracket n, k, d \rrbracket$ (the witness is a $\llbracket 12, 8 \rrbracket$ toy).

6.2 A certified solver-free distance ceiling on odd lattices

Corollary J-G1 says the collapse channel is empty when ℓ, m are both odd, which raises the design question of what actually lives in the collapse-free region. Answering it needed one new tool and one correction to our own bookkeeping.

Already known (reproduced, not claimed). For ℓ, m odd, $|G|$ is odd, so $R = \mathbb{F}_2[G]$ is semisimple by Maschke and splits as $\prod_{\chi} \mathbb{F}_{\chi}$ over Frobenius orbits of characters. Multiplication by a is diagonal, hence $\text{Ann}(a) = \bigoplus_{a(\chi)=0} \mathbb{F}_{\chi}$ and $k = 2 \dim_{\mathbb{F}_2} I = 2 \sum_{\chi: a(\chi)=b(\chi)=0} [\mathbb{F}_{\chi} : \mathbb{F}_2]$: the degree-weighted common-root law. This appears as $k = 2 \deg \gcd(a, b, x^{\ell} - 1)$ in Panteleev–Kalachev (arXiv:1904.02703, Prop. 1), in Lin–Pryadko (arXiv:2306.16400, Eq. 47), as Eq. (11) of Wang–Mueller (arXiv:2408.10001v4) for coprime lattices, as Thm. 2.6 of Postema–Kokkelmans (arXiv:2502.17052v4), and as Cor. 2.11–2.12 of Eberhardt–Steffan (arXiv:2407.03973v1). We use it; we do not claim it.

Theorem K (reciprocal-pole logical isomorphism; bar-aware form of the published principal-code structure). The coefficient ideal $I = \text{Ann}_{\text{left}}(a, b)$ is what `nullspace(HX.T)` computes; it is not itself the physical right kernel. Put $J = \bar{I}$ and $\mathcal{P} = J \oplus J$. Then

$$\mathcal{P} \subseteq \ker H_X, \quad \mathcal{P} \cap S_Z = 0, \quad \dim \mathcal{P} = k,$$

so $\mathcal{P} \cong \ker H_X / S_Z$. Indeed, I is supported on the common-zero set Z and J on Z^{-1} ; physical column multiplication evaluates the reciprocal character, while $S_Z = \{(\lambda \bar{b}, \lambda \bar{a})\}$ vanishes on Z^{-1} . The rank statement then follows from $k = 2 \dim I$. $\square$

This is the Eberhardt–Steffan principal-code isomorphism in the repository’s matrix convention. It gives

$$d(P) = \min_{0 \neq p \in \mathcal{P}} \min_{s \in S_Z} \text{wt}(p + s), \quad d(P) \leq d_{\text{pole}} := \min_{0 \neq u \in J} \text{wt}(u).$$

The raw pole ceiling costs $O(2^{k/2} \ell m)$ bit operations: microseconds for $k \leq 32$, no solver, decoder, or sampling. Reducing selected pole vectors modulo S_Z under several information-set orders gives tighter self-certifying witnesses; randomisation affects tightness, never soundness. For $[[90, 8, 10]]$ the raw ceiling is 20 and the reduced witness has weight exactly 10.

The bar is load-bearing. On $(7, 7)$ $[[98, 6, 12]]$ every basis vector of raw I fails $H_X(u, 0)^T = 0$, while every basis vector of $J = \bar{I}$ passes. An intermediate draft used raw I in a fallback; the regression caught it before the final PDF and the route is retracted. The earlier $I \cap \bar{I}$ statement was sound but unnecessarily weak.

Validation, and a correction. A read-only novelty check found the pole isomorphism already published and did not find the explicit general-BB minimum-pole-weight ceiling or its use as an exhaustive rejection oracle; the latter is nevertheless a short corollary and is not oversold as a deep new theorem. The same check refuted a claim that $[[90, 8, 10]]$ on $(15, 3)$ is the only odd $\times$ odd BB instance in print. We sourced 29 from arXiv:2308.07915, arXiv:2407.03973v1, arXiv:2408.10001v4, arXiv:2502.17052v4, and Liang et al. [3]. The later audit also missed Liang et al.’s two exact $[[234, 8, 18]]$ Table III rows; that omission is corrected. Our statement was correct only for the seven entries in our baseline table; literature-wide it was wrong and is withdrawn.

On these 29 instances the reciprocal-pole isomorphism passes 29/29 rank audits. Three independent k routes agree internally on all 29 and with printed k on 27; two transcribed Wang–Mueller App. C rows, $(5, 9)$ and $(7, 11)$, give internal $k = 0$ against printed 4 and 6, and are excluded. All 27 reproduced rows satisfy “pole ceiling $\geq$ reported d ”, but decoder-reported values remain only a sanity check: Wang–Mueller uses BP-OSD `distance_upperbound` and Postema–Kokkelmans labels table values Monte-Carlo estimates. Eight rows are independently exact-certified here: the previous six plus Liang et al.’s two $[[234, 8, 18]]$ rows locally replayed by EXP-067. There are zero violations, with slack min/median/max 2/22/34. The screen uses only independent two-sided certificates as domination thresholds; no estimate or un-replayed source value can reject a candidate.

Exhaustive census. Over all 65 odd lattices with $\ell m \leq 180$ ($n \leq 360$) we enumerated every weight- ≤ 3 pair — 4.23×10^9 pairs — computing k by the ideal route with a matrix-route cross-check and **zero mismatches**, and re-verifying the J-G1 mechanism directly (273 idempotence tests, zero violations) on lattices outside our catalogue’s $m \in \{3, 6\}$. Grouping polynomials by Ann makes the pass exact rather than sampled, since k and the ceiling depend only on the pair of annihilators.

Exact discoveries before fixed-point closure. The first exact screen waves produced three connected, row-space-indecomposable BB codes:

(ℓ, m)	A	B	$[[n, k, d]]$
$(5, 3)$	$1 + x + x^3y$	$1 + xy + x^4y$	$[[30, 8, 4]]$
$(9, 3)$	$1 + y + x^3y^2$	$1 + x + x^2$	$[[54, 8, 6]]$
$(7, 9)$	$1 + \pi + \pi^{58}$	$\pi^3 + \pi^{16} + \pi^{44}$	$[[126, 12, 10]]$

All sector distances are independently CP-SAT exact with explicit witnesses. A primary-table search found no matching $[[30, 8, 4]]$ BB constructor, but Grassl’s table contains an explicit general stabilizer $[[30, 8, 7]]$: the BB constructor is new within the checked BB corpus but globally dominated. The $[[54, 8, 6]]$ and $[[126, 12, 10]]$ results exactify Wang–Mueller BP-OSD parameters (the latter costs 306s). All three become hash-bound local references for the fixed-point screen.

Orbit-class exactification at $n = 162$ (EXP-056). For the Wang–Mueller $(3, 27)$ code $A = 1 + y^{10} + y^{14}$, $B = y^{12} + x + x^2$, the 255 nonzero logical classes form 20 orbits under the verified order-162 translation/reflection group. Each representative is encoded as an affine coset of the 77-dimensional Z -stabilizer rowspace. Every H_X column has odd degree three, so all kernel words have even weight and excluding weight ≤ 12 also excludes weight 13. Kissat proves all 20 class CNFs UNSAT and repeats all 20 on fresh digest-bound replay; an explicit weight-14 word passes independent NumPy and bitset checks. The exact BB duality permutation gives $d_X = d_Z = 14$, promoting the source BP-OSD estimate to a local two-sided certificate. Generic fixed-functional sectors do not inherit the monolithic origin anchor; a synthetic counterexample caught and repaired that latent alternative-mode bug (FR-027).

Adaptive frontier ratchet and CRT transport (EXP-057–065). Above $n = 162$, constructor-bound monolithic certificates exactify $[[170, 16, 10]]$ and seven distinct $[[186, 10, 14]]$ symmetry classes. At block size 105, an explicit CRT coordinate permutation proves that the coprime lattices $(15, 7)$, $(21, 5)$ and $(35, 3)$ are one BB search space. A new adaptive method uses the all-odd column-degree parity law, exact BB $d_X = d_Z$ duality, and one-sector raw-kissat decisions at even caps only. It exactifies thirteen $[[210, 18, 8]]$ classes, $[[210, 24, 4]]$, $[[210, 14, 12]]$ and two $[[210, 10, 16]]$ classes; promoted representatives are replayed and hash-bound. A separate $k = 8$ residual has independent weight-16 witnesses from CaDiCaL and Maple and is dominated at threshold 16; no lower-bound or exact-distance claim is made for it. Two independent alternatives fail closed: exact affine-trellis widths are 48–99 on the hard types (above the 5,000,000-state gate), while the $H = 7$ orbit projection gives only the rigorous lower bound 4. A pole-stabilizer audit confirms $H = 5$ on all thirteen easy classes but falsifies “hard implies $H = 1$ ”.

Fixed-point Pareto screen (complete through $n = 210$). Promoting every new fixed point into the constructor- and certificate-hash-bound reference set, and replacing two duplicate $n = 210$ solves by exact CRT transport, closes all 20 odd lattices with nonempty weight-3 frontier through

$n = 210$, $8 \leq k \leq 24$. Exact quotienting leaves 4,020 classes representing 122,833 translation-normalised pairs. Of these, 3,816 have an admissible independently exact reference: all are dominated (3,364 by reduced-pole logical witnesses, 395 by bounded or adaptive CDCL witnesses, and 57 by the exact CP-SAT fallback). All 3,759 persisted witnesses are checked in $\ker H_X \setminus S_Z$. The other 204 high- k classes have no certified reference and are labelled `no_reference`, not dominated. There are zero survivors and zero undecided among referenced classes. Direct shards bind the census and reference hashes; monotone rebinding archives prior shards and rechecks every carried witness, while transported shards carry exact coordinate-permutation proofs.

Exact $n = 234$ closure (EXP-066/067). The two next frontiers are exhaustively represented. On (13, 9) all 84 classes are dominated at the exact $k = 12$, $d = 12$ threshold. On (39, 3), EXP-066 enumerates all $12|\text{GL}(2, 3)| = 576$ automorphisms from $\mathbb{Z}_{39} \times \mathbb{Z}_3 \cong \mathbb{Z}_{13} \times \mathbb{F}_3^2$, checks every coordinate map as a bijective homomorphism, and compresses 182 hard classes to 30 exact bundles. Matrix-verified witness transport closes 28 bundles / 158 classes.

The final two bundles are Liang et al.’s two published [234, 8, 18] Table III rows. EXP-067 enumerates every quotient-lattice generator image, finds one map per row, and verifies the mapped H_X, H_Z rowspaces by explicit qubit permutation. A minimum logical has no proper zero-syndrome subset, so connected-cluster enumeration may be rooted at its least coordinate. Translation plus an exact block-column swap reduces the lower proof to two roots per code. Each root exhausts weights 1–16 and is replayed; the even-kernel law excludes 17, physical weight-18 witnesses pass independent NumPy/bitset checks, and BB duality gives $d_X = d_Z = 18$.

Promoting that exact reference closes the 24 former undecided classes by rechecked weight-18 witnesses. Across 22 lattices there are 4,862 classes / 150,581 represented pairs; all 4,658/4,658 referenced classes are dominated, 204 are `no_reference`, and there are zero survivors or undecided. The 4,658 dominations comprise 4,048 direct physical witnesses, 158 full-automorphism transports, 395 bounded CDCL witnesses, and 57 exact CP-SAT fallbacks. Exact fixed-point closure reaches $n = 234$; the algebraic k census remains complete through $n = 360$.

Final proof-surface review found and closed a separate aggregation weakness: the monotone rebinder had preserved old `dominated*` labels while non-target shards received only aggregate validation, and the 57 CP-SAT fallback rows stored `d_found` without support. EXP-068 recovers 29 independent physical Z -logicals, transports 28 by exact CRT permutations, and hash-binds all 57. Validator v11 now rebuilds every class identity or transport cover, threshold/source and physical witness on all 22 shards; rebinding validates pre-state, post-state and archive, and EXP-066 invokes the same global validator.

The multithreaded dist-m4ri coordinator has a one-root completion race, so those runs were cancelled. Certificates use the source-archived, hash-bound legacy single-thread entrypoint, which calls the recursion directly and emits terminal -16 only after full exhaustion. It remains native trusted code rather than a proof-trace checker; replay and the published exact result are redundant evidence, not formal verification of the executable.

7 The exact trade law, its residual, and remaining conjectures

Theorem I (restated; proved in §3.2). *If $T(P) \geq k_P/2$ and $d_Q > d_Z(P)$, then $k_Q = k_P - T(P)$ exactly. This covers 138 of 139 certified parents and all 57 family-closed ones at $n \leq 144$; the saturation observed in §5.2 and the 496/496 small-lattice increases are instances of it.*

Three layers of machine evidence now sit beneath the theorem and delimit its residual:

1. *Catalogue upper law.* On all 284 classified rows, $\dim \bar{\Delta} \leq T(P)$ with zero exceptions (spread of $\dim \bar{\Delta} - T$ only non-positive, down to -24). For $T \geq k_P/2$ parents this is Lemma 3; the law holding also on the one $T < k_P/2$ parent's rows is evidence the residual class is benign.
2. *Small-lattice probe (EXP-040).* 64 parents across four lattices, 26,898 unseen $\delta > 0$ perturbations: $\dim \bar{\Delta} < T$ on 23,634, $= T$ on 3,264, $> T$ on **zero**; all 496 strict increases at $= T$, with actual containment $\bar{M} = \bar{\Delta}$ verified, not just dimensions. All probed parents had $T \in \{k_P/2, k_P\}$, so these are theorem-forced.
3. *Reversals.* All 7 catalogue reversals at $\dim \bar{\Delta} = T = k_P/2$ with $\bar{M} = \bar{\Delta}$.

Conjecture B' (residual saturation — decided on the known witness). *For parents with $T < k_P/2$ (one known: 9a7638586033, where increase sandwiches to $4 \leq \dim \bar{\Delta} \leq 6$), distance increase still forces $\dim \bar{\Delta} = T$.*

On the only known residual parent the question is now *decided*, and the answer is stronger than saturation: increase over 9a7638586033 is *impossible*, because the dressing rows of every valid perturbation live in a fixed two-dimensional space (EXP-045): over its entire valid family ($\dim V = 112$, 2,533,006,645 instances enumerated exhaustively through weight 6, per-weight counts matching $\binom{112}{k}$ exactly), $\dim \bar{\Delta} \in \{0, 2\} \subset [0, 2] < 4 = T$, so the Theorem-H premise is never met and $d_Q \leq d_Z(P) = 6$ for *every* valid $[C \ D]$ of that parent. The sandwich $[4, 6]$ is never attained. Whether the residual class contains any other parent at larger lattices ($\ell \cdot m > 40$) remains open (§6); on the known witness the k -halving confinement picture is therefore complete with no exceptional direction.

Conjecture C (k -halving confinement). *Any perturbation with $d_Q > d_Z(P)$ has $k_Q/k_P = 1/2$. On every $T = k_P/2$ parent this is immediate from Theorem I; the conjecture is residual exactly on $T < k_P/2$ parents and on the 63 uncertified ones ($n \in \{180, 360\}$). Every one of the 284 classified rows with $k_Q/k_P \in \{1, 3/4, 5/6\}$ is capped a priori (§5.1, zero exceptions).*

Conjecture D (distance parity). *Certified reversals gain exactly $+2$ in distance.*

Evidence: the uniform gap $d_Q - d_Z(P) = 2$ across all 7 reversals, whose exact d_Q values are certified; consistent with d_Z even throughout. (The probe stores only the increase decision, not the achieved d_Q , so its 496 increases are not parity evidence.)

The open problem that remains is not *whether* perturbation pays — it pays exactly T — but *whether anything on the CSS side of the envelope can ever be beaten this way*: §6's domination table says no for all seven known increases.

8 Reproduction and provenance

Environment: macOS Darwin 25.5.0 arm64, Apple M3 Ultra, 28 logical CPUs, 96 GiB; Python 3.13.9 in .venv; numpy 2.4.6, scipy 1.18.0, stim 1.16.0, pymatching 2.4.0, sinter 1.16.0, ldpc 2.4.1, galois 0.4.11, ortools 9.15.6755, python-sat 1.9.dev13; solvers CaDiCaL 1.9.5 and Kissat 4.0.4 via PySAT. Full suite: 1007 passing tests, 1 skipped (1008 collected).

claim	artifact	experiment
Theorem G, machine-checked on catalogue rows	src/qec_research/codes/pbb_survival.py, tests/test_pbb_survival.py (15 checks)	EXP-038

Theorem H, module + exact T enumeration	src/qec_research/codes/pbb_nogo.py, tests/test_pbb_nogo.py (17 checks)	EXP-039
Gross closure, 63 closed parents, 253 rows capped, complete at $n \leq 144$	results/partial_runs/exp039_nogo_module.json	EXP-039
Saturation table, 7/7 non-vacuous, gate	exp039_nogo_module.py gate	EXP-039
Certified d_Q lower bounds for reversals	results/partial_runs/exp037/row_*.json	EXP-037
Theorem I legs: $k_P = 2 \dim L$ (202 parents), $\dim \bar{\Delta} \leq k_P/2$ (368 rows), scope exception	tests/test_pbb_theorems.py, results/processed/exp040_saturation_probe.json	EXP-039/040
All 7 reversals CSS-dominated at equal n (162 exact candidates)	results/processed/exp036_envelope_check.json	EXP-036/037
5 certified reversals, 41 dominations of 87	results/partial_runs/exp036_delta_closure.json	EXP-036
Exact $d = 12$ for 12_6_0193, two-sided	results/certificates/pbb_12_6_0193_distance.json	EXP-035
Genuinely non-CSS (12_6_0193)	notes/novelty_matrix.md, EXP-034 records	EXP-034
Catalogue-wide CSS-envelope classification	results/partial_runs/exp037_envelope_classification.json	EXP-037
Strict-provenance $\delta > 0$ audit (155 rows)	results/processed/exp027_delta_audit.json	EXP-027
Translation symmetry break soundness, $6.2\times$	tests/test_translation_symmetry_break.py	FR-022
DEM mechanism-distance intervals, LER, latency	reports/technical_report.md §5-6	EXP-016/029
Theorem J-C: $D=0$ collapse family decidable, 192/202 parents	results/processed/exp046_generic_syzygy.json	EXP-046
Theorem J-E immunity decision, 10 immune exact	results/processed/exp047_exact_demotion_decision.json	EXP-047
Exact collapse census: 133/133 terminal-UNSAT, 99 strict drops (98 full- k)	results/processed/exp048_exact_collapse_sweep.json	EXP-048
Demote-fraction enumeration, 196 parents ($k_P \leq 20$, 2,693,100 classes)	results/processed/exp050_demote_fractions.json, results/partial_runs/exp050/fraction_*.json	EXP-050
X-U distance certificates $d_P = 10$ + family-closure ($T = 8 = k_P$ respectively $T = 16 = k_P$), phase2_109 $d_P = 6$, $T = 4$	results/partial_runs/exp039/parent_e15b3d6480981d1e.json, results/partial_runs/exp039/parent_765498fc67b2b4b9.json, results/partial_runs/exp039/parent_dc98a5e4c350a656.json	EXP-039/049
X-monotone channel exclusion on the 10 immune parents: 8 INFEASIBLE certificates + 2 weight-8 witnesses	results/processed/exp051_xmonotone_lightscan.json, tests/test_exp051_xmonotone_lightscan.py (3 checks)	EXP-051
Final X-side partition 192/8/2 (= 202)	jointly derived: EXP-047 demote/immune decision + EXP-049 dedicated certificates + EXP-051 exclusions	EXP-047/049/051
Theorem J-E'': demote fixed set $= I^\infty M$; 202/202 fractions $\in \{0, 1\}$ (192/10, 0 mixed), chains ≤ 3 , 196 integer crosschecks	results/processed/exp052_ideal_power_trichotomy.json, tests/test_exp052_ideal_power_trichotomy.py (11 checks)	EXP-052
Theorems J-G/J-H/J-I/J-J: ideal invariance ($\dim S = 2 \dim I^\infty$, 202/202 vs EXP-052), $I^2 = 0$ on all 192 demoting / $I^2 = I$ on all 10 immune, odd-lattice corollary (3,600 parents), baseline table ($\llbracket 90, 8, 10 \rrbracket$ immune), coset criterion exact (10 vs 192), lemma battery, mixed witness (3 routes)	results/processed/exp053_ideal_classification.json, tests/test_exp053_ideal_invariant.py (13 checks)	EXP-053
Weight- ≤ 3 mixed census: 653,022,021 pairs over 18 lattices, zero mixed, 450 independent cross-checks	results/processed/exp054_mixed_census.json	EXP-054

Odd-lattice census: 65 lattices, 4,229,823,962 pairs, zero k mismatches	results/processed/exp055_odd_lattice_sweep. json	EXP-055
Exact frontier ratchet: $\llbracket 170, 16, 10 \rrbracket$, seven $\llbracket 186, 10, 14 \rrbracket$, thirteen $\llbracket 210, 18, 8 \rrbracket$, $\llbracket 210, 24, 4 \rrbracket$, $\llbracket 210, 14, 12 \rrbracket$, and two $\llbracket 210, 10, 16 \rrbracket$; independent weight-16 witnesses (not exactness) for the final $k = 8$ residual; exact CRT transport	results/certificates/exp060_210_18_8_ distance.json, results/processed/exp059_ coprime_transport.json	EXP-057-065
Historical exact closure through $n = 210$: 4,020 classes / 122,833 pairs, 3,816/3,816 referenced dominated	results/processed/exp055_odd_lattice_screen. json	EXP-055-065
Exact closure through $n = 234$: 22 lattices / 4,862 classes / 150,581 pairs, 4,658/4,658 referenced dominated, 204 no_reference , zero survivor/undecided; 576-map audit, two replayed rooted certificates for Liang et al.'s $\llbracket 234, 8, 18 \rrbracket$ rows, and physical witness bindings for all 57 fallback dominations	results/processed/exp066_n234_frontier.json, results/certificates/exp067_234_8_18_ bundle19_distance.json, results/certificates/ exp067_234_8_18_bundle22_distance.json, results/partial_runs/exp068_screen_witnesses/	EXP-066-068

Every SAT decision records its canonical CNF SHA-256 and encoding version; verdicts are re-derived from rebuilt matrices on replay, and stamps that fail to hash-bind are rejected rather than trusted. Forgery regression tests confirm that crude tampering dies arithmetically or under replay.

Reproduce the headline result:

```
cd math/qec
PYTHONPATH=src .venv/bin/python experiments/exp039_nogo_module.py run --ns 144
PYTHONPATH=src .venv/bin/python experiments/exp039_nogo_module.py gate
PYTHONPATH=src .venv/bin/python experiments/exp055_odd_lattice_sweep.py run \
  --max-dim 180 --workers 24 --force
PYTHONPATH=src .venv/bin/python experiments/exp056_odd_distance.py \
  run-classes --indexes all --force
PYTHONPATH=src .venv/bin/python experiments/exp056_odd_distance.py \
  run-classes --indexes all --replay
PYTHONPATH=src .venv/bin/python experiments/exp056_odd_distance.py assemble
PYTHONPATH=src .venv/bin/python experiments/exp055_odd_lattice_sweep.py \
  literature
PYTHONPATH=src .venv/bin/python experiments/exp055_odd_lattice_sweep.py screen \
  --lattices 3x3,5x3,7x3,9x3,9x5,15x3,7x7,21x3,9x7,15x5,25x3,9x9,27x3 \
  --workers 8 --time-limit 120 --force
PYTHONPATH=src .venv/bin/python experiments/exp055_odd_lattice_sweep.py \
  screen-certify
PYTHONPATH=src .venv/bin/python -m pytest \
  tests/test_exp055_odd_lattice.py tests/test_exp056_odd_distance.py -q
```

9 Related work and novelty

Bivariate-bicycle codes and the Gross code are due to Bravyi et al. [1]. The PBB construction and the 368-row catalogue are from [2]. Cruz-Benito et al. [2] (v1, Sec. III.2, Lemma 0) give the validity condition; we derive it independently from the binary symplectic Gram matrix and confirm the correct condition is *symmetry* of $AC^T + BD^T$, not its vanishing. Multivariate bicycle generalisations are in [5]; existence and characterisation results for BB codes in [4]. ASC [8] certifies no-depth-6 for IBM BB codes; our depth criterion is independent and basis-independent.

To our knowledge the following are new here: (a) the identification of Δ as an R -submodule and the resulting orbit-absorption argument; (b) the exact dimension identity $k_Q = k_P - \dim \bar{\Delta}$

as a certified structural fact rather than an observed coincidence; (c) the perturbation-independent no-go bound (Theorem H) and its exact T enumeration with a terminating UNSAT certificate; (d) the closure of the entire perturbation family over the Gross code; (e) the saturation observation $k_Q = k_P - T$ on all certified reversals; and (f) the proved confinement of distance increase to a k -halving class.

10 Conclusion

The PBB construction’s extra freedom is real but not free. It is governed by a single submodule Δ whose dimension is exactly the dimension deficit, and whose module structure forces any distance increase to absorb entire translation orbits of the parent’s minimum-weight logicals. This yields a computable, perturbation-independent obstruction that closes 63 of 202 published parents outright — the Gross code among them — caps 253 of 368 catalogue rows with no search, confines distance increase to a k -halving class, and is saturated with slack exactly zero on all 7 certified distance-increasing perturbations known to us.

The abstract $\llbracket n, k, d \rrbracket$ route past the CSS bivariate-bicycle envelope is, over these parents, closed by theorem. What remains genuinely open is the circuit-level question: whether a verified fault-tolerant syndrome-extraction circuit for a non-CSS PBB can beat the Gross code end-to-end under identical noise. Nothing in our circuit-level evidence so far favours it.

References

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- [2] Juan Cruz-Benito, Andrew W. Cross, David Kremer, and Ismael Faro. Evolutionary discovery of bivariate bicycle codes with LLM-guided search, 2026. [arXiv:2606.02418](#).
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- [7] Fuchuan Wei, Zhengyi Han, Austin Yubo He, Zimu Li, and Zi-Wen Liu. Theory of low-weight quantum codes, 2026. [arXiv:2601.19848](#).
- [8] Kai Zhang, Dingchao Gao, Zhaohui Yang, Runshi Zhou, Fangming Liu, Zhengfeng Ji, and Jianxin Chen. Optimal compilation of syndrome extraction circuits for general quantum LDPC codes, 2026. [arXiv:2603.21499](#).